

SOULAYMA HRIRA

FINAL-YEAR COMPUTER ENGINEERING STUDENT |

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SUMMARY

Final-year engineering student with practical experience in machine learning, data analysis, and web development. Skilled in building ML models, processing large datasets, and collaborative coding. Eager to apply these skills to high-performance computing and scientific data challenges.

EXPERIENCE

T24 BANKING DATA ENGINEERING INTERNSHIP

Biat ITT — jul 2026– sep 2026

- Designed a star schema using synthetic T24 banking data.
- Built ETL/ELT pipelines using Python, SQL, and PySpark for banking data transformation.
- Applied Medallion Architecture (Bronze, Silver, Gold) with Delta Lake and Delta Tables.
- Used dbt for SQL-based data transformations and modular data modeling.
- Explored MLflow for machine learning experiment tracking.
- Tech Stack: Databricks, PySpark, Delta Lake, Delta Tables, dbt, MLflow, Python, SQL, ETL/ELT, Medallion Architecture.

AI Engineering Intern — Remote

Smartovate — Feb 2026– Mar 2026

- Participated in the development of a RAG-based educational assistant deployed on Microsoft Azure.
- Gained hands-on exposure to Azure AI services, embeddings, and vector search concepts in a production-like environment.
- Tech Stack: Azure OpenAI, Azure AI Search, Python, RAG systems

AI-Powered Educational Platform Development

Pedago.ai — jun 2025– oct 2025

- Worked on AI-powered educational tools designed for teachers and students.
- Developed and integrated optimized AI prompts and implemented advanced LLM-based APIs to enhance the learning experience.
- Contributed to full-stack development and system architecture of the platform.
- Gained hands-on experience in designing, building, and deploying AI-driven educational solutions.
- tech stack : Spring Boot, Angular, H2 Database, Formily, Swagger, AI/LLM APIs, Prompt Engineering

EDUCATION

Engineering in Software Engineering and Business Intelligence — Sfax, Tunisia

International Institute of Technology — Sep 2024– Jun 2027

Bachelor's in compuer science and big data — Sfax, Tunisia

Higher Institute of Computer Science and Multimedia — Sep 2024– Jun 2027

PROJECTS

AdventureWorks BI Pipeline
Academic Project

- Built an end-to-end BI pipeline simulating a real-world data warehousing workflow using the AdventureWorks dataset.
- Designed data models and implemented ETL processes to extract, transform, and load data into a structured analytical environment
- Developed analytical models and interactive dashboards to deliver key business KPIs and insights
- Tech stack: SQL Server, SSMS, SSIS, SSAS, Power BI

GitHub Repo

AI SYSTEM FOR MEDICAL ANOMALY DETECTION AND REPORT GENERATION
Academic Project

- Preprocessed medical images in DICOM format (normalization, photometric correction) using Python, pydicom and NumPy
- Detected and segmented radiological anomalies (tumors, nodules, fractures)
- Classified the severity of detected anomalies

Teck stack:Python, pyTorch, YOLOv8py , NumPy , vLM , Google Colab

INSURANCE CLAIMS PREDICTION
Academic Project

GitHub Repo

- Built an end-to-end ML pipeline for predicting home insurance claim probability from a noisy dataset
- Performed data preprocessing, feature engineering, and trained multiple supervised models to evaluate predictive performance
- Assessed models using standard metrics and learning curves, focusing on error analysis and interpretability of key risk factors
- Tech stack: Python, Pandas, Scikit-learn, Matplotlib/Seaborn, Git

CERTIFICATIONS

- Microsoft Certified: AI-102 – Designing and Implementing an Azure AI Solution
- AI Engineer for Data Scientists Associate-Datacamp